Learning Objectives 7.1 and 7.2: I can use the Biot-Savart law to simple magnetostatic systems to find the magnetic field at an observation point. I can articulate the meaning of both the integral form and point form of Gauss's Law for magnetism.

Show your work. Take $\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **A finite wire.** A straight wire on the z axis carries $I = 2$ A in the $+z$ direction from $z = -10$ cm to $z = 10$ cm.

   (a) Starting from the Biot-Savart law, derive the field at a distance ρ from the wire in the plane $z = 0$.

   With $d\vec{l} = \hat{\mathbf{z}}\, dz'$ and $\vec{R} = \rho\hat{\boldsymbol{\rho}} - z'\hat{\mathbf{z}}$, the cross product is $d\vec{l} \times \vec{R} = \hat{\phi}\, \rho\, dz'$, so

   $$\vec{H} = \int_{-L/2}^{L/2} \frac{I\, \hat{\phi}\, \rho\, dz'}{4\pi\left(\rho^2 + z'^2\right)^{3/2}} = \hat{\phi}\frac{I\rho}{4\pi}\left[\frac{z'}{\rho^2\sqrt{\rho^2 + z'^2}}\right]_{-L/2}^{L/2}$$
   $$= \hat{\phi}\frac{IL}{4\pi\rho\sqrt{\rho^2 + L^2/4}}$$

   The integral follows from $z' = \rho \tan\alpha$, as in Lesson 18.

   (b) Find $\vec{H}$ and $|\vec{B}|$ at the point $(5, 0, 0)$ cm.

   $$H = \frac{(2)(0.2)}{4\pi(0.05)\sqrt{0.05^2 + 0.1^2}} = 5.69 \text{ A/m}$$
   $$\vec{H} = 5.69\, \hat{\mathbf{y}} \text{ A/m}$$
   $$B = \mu_0 H = 7.16\, \mu\text{T}$$

   $\vec{H}$ = $5.69\, \hat{\mathbf{y}}$ A/m

   $|\vec{B}|$ = $7.16\, \mu$T

   At $(5, 0, 0)$ cm, $\hat{\phi} = \hat{\mathbf{y}}$.

   (c) Compare your answer with the field of an infinite wire carrying the same current. What fraction of the infinite-wire field does the finite wire give?

   $$H_\infty = \frac{I}{2\pi\rho} = \frac{2}{2\pi(0.05)} = 6.37 \text{ A/m}$$
   $$\frac{H}{H_\infty} = \frac{5.69}{6.37} = 0.894$$

   fraction = 0.894

2. **A loop on its axis.** A circular loop of radius $a = 4$ cm lies in the plane $z = 0$, centered on the z axis, and carries $I = 0.5$ A counterclockwise seen from above.

   (a) Starting from the Biot-Savart law, derive the field at height z on the axis. Explain why the radial part of the integral is zero.

   With $d\vec{l} = \hat{\boldsymbol{\phi}}'\, a\, d\phi'$ and $\vec{R} = z\hat{\mathbf{z}} - a\hat{\boldsymbol{\rho}}'$, $R = \sqrt{a^2 + z^2}$ for every element and $d\vec{l} \times \vec{R} = a\, d\phi' \left(z\hat{\boldsymbol{\rho}}' + a\hat{\mathbf{z}}\right)$:

   $$\vec{H} = \frac{Ia}{4\pi \left(a^2 + z^2\right)^{3/2}} \int_0^{2\pi} \left(z\hat{\boldsymbol{\rho}}' + a\hat{\mathbf{z}}\right) d\phi' = \hat{\mathbf{z}} \frac{Ia^2}{2\left(a^2 + z^2\right)^{3/2}}$$

   The unit vector $\hat{\boldsymbol{\rho}}'$ turns with ϕ', and its integral around a full circle is zero: the elements on opposite sides of the loop produce equal and opposite radial fields.

   (b) Find $|\vec{H}|$ at the center of the loop and at $z = 3$ cm.

   $$H(0) = \frac{I}{2a} = \frac{0.5}{2(0.04)} = 6.25 \text{ A/m}$$

   $$H(3 \text{ cm}) = \frac{(0.5)(0.04)^2}{2\left(0.04^2 + 0.03^2\right)^{3/2}} = 3.20 \text{ A/m}$$

   center = 6.25 A/m

   $z = 3$ cm: 3.20 A/m

   (c) At what height on the axis is the field half its value at the center?

   $$\left(\frac{a^2}{a^2 + z^2}\right)^{3/2} = \frac{1}{2} \implies a^2 + z^2 = 2^{2/3} a^2$$

   $$z = a\sqrt{2^{2/3} - 1} = 0.766\, a = 3.07 \text{ cm}$$

   $z =$ 3.07 cm

3. **A square loop.** A square loop of side $s = 8$ cm carries $I = 2$ A.

   (a) Use the finite-wire result to find the field at the center of the loop.

   $H =$ 22.5 A/m

   The center is on the bisecting plane of each side, with $L = s$ and $\rho = s/2$, and all four sides produce fields in the same direction:

$$H_{\text{side}} = \frac{Is}{4\pi(s/2)\sqrt{s^2/4 + s^2/4}} = \frac{\sqrt{2}\,I}{2\pi s} = 5.63 \text{ A/m}$$

$$H = 4H_{\text{side}} = \frac{2\sqrt{2}\,I}{\pi s} = 22.5 \text{ A/m}$$

   (b) A circular loop of radius $a = s/2$ fits inside the square and carries the same current. Find its field at the center, and explain why it is larger than the square's even though the square has more wire.

$$H = \frac{I}{2a} = \frac{2}{2(0.04)} = 25.0 \text{ A/m}$$

   Every element of the circle is 4 cm from the center and broadside to it. The wire near the corners of the square is farther away, up to $s/\sqrt{2} = 5.66$ cm, and sees the center at a smaller $\sin\theta$, so it contributes less.

4. **Gauss's law for magnetism.**

(a) Which of these fields could be a magnetic flux density? Justify each answer.

$$\vec{B}_1 = x\hat{\mathbf{x}} - y\hat{\mathbf{y}}, \qquad \vec{B}_2 = y\hat{\mathbf{x}} + x\hat{\mathbf{y}}, \qquad \vec{B}_3 = x\hat{\mathbf{x}} + y\hat{\mathbf{y}}$$

$\nabla \cdot \vec{B}_1 = 1 - 1 = 0$ possible

$\nabla \cdot \vec{B}_2 = 0 + 0 = 0$ possible

$\nabla \cdot \vec{B}_3 = 1 + 1 = 2$ not possible: a magnetic field has $\nabla \cdot \vec{B} = 0$

(b) The Earth's field at a site is $50\ \mu$T, pointing straight down. Find the magnetic flux through a hemispherical dome of radius 1 m sitting on flat ground. Use Gauss's law to avoid integrating over the dome.

$\Phi =$ $157\ \mu$Wb

The dome and the flat disk under it form a closed surface, so the flux through the dome equals the flux through the disk:

$$\Phi = B\pi R^2 = \left(50 \times 10^{-6}\right)\pi\,(1)^2 = 157\ \mu\text{Wb}$$

5. **The force on a moving charge.** A long straight wire on the z axis carries 10 A in the $+z$ direction. An electron passes the point $(1, 0, 0)$ cm moving at 10^6 m/s in the $+z$ direction.

   (a) Find $\vec{B}$ at the electron.

$$\vec{B} = \hat{\phi}\frac{\mu_0 I}{2\pi\rho} = \hat{\mathbf{y}}\frac{\left(4\pi \times 10^{-7}\right)(10)}{2\pi(0.01)} = 200\,\hat{\mathbf{y}}\,\mu\text{T}$$

$\vec{B}$ = $200\,\hat{\mathbf{y}}\,\mu$T

   (b) Find the force on the electron, and state whether it pulls the electron toward the wire or pushes it away.

$$\vec{F} = q\vec{v} \times \vec{B} = \left(-1.602 \times 10^{-19}\right)\left(10^6\hat{\mathbf{z}}\right) \times \left(2 \times 10^{-4}\hat{\mathbf{y}}\right)$$
$$= \left(-3.20 \times 10^{-17}\right)(-\hat{\mathbf{x}}) = 3.20 \times 10^{-17}\,\hat{\mathbf{x}}\ \text{N}$$

$\vec{F}$ = $3.20 \times 10^{-17}\,\hat{\mathbf{x}}$ N

The force points away from the wire. The electron's motion is a current opposite to the wire's, and opposite currents repel.

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